

Protective Devices

Fuses

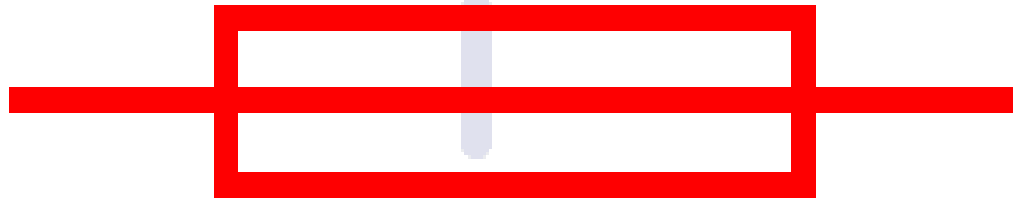
Fuses

- **Introduction:**
- A **fuse** is an **electrical** / **electronic** or **mechanical** safety device designed to protect circuits from overcurrent and overload conditions. Invented by Thomas Alva Edison in 1890, fuses are used widely to safeguard electrical systems during electrical faults.
- A fuse is a device used to protect the circuits and equipment against overload and short circuits.
- Fuse element materials with a low melting point and high conductivity.

- Fuse operation is based on the heating effect of the current when it flows through the element. In normal operation, the heat is dissipated rapidly into the surrounding air, and the element remains at below its melting point. When a fault occurs in the electric circuit, the current exceeds the limiting value and cannot be dissipated heat fast enough; as a result, the element melts and breaks the circuit.
- Fuse has inverse time current characteristics.
- Its rating can start from a few mA to several kA.
- It has many forms and shapes depending on its application.

While various types of fuses exist, their fundamental function remains the same: to interrupt the flow of current when it exceeds a predetermined limit. In today's explanatory article for protective devices, we will discuss different types of fuses, their construction, operating principles, and applications within various electrical and electronic systems.

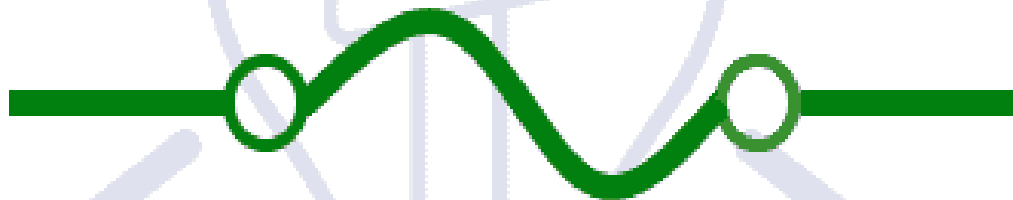
IEEE/ANSI Standard



IEC Standard



IEEE/ANSI Standard



Old Symbol

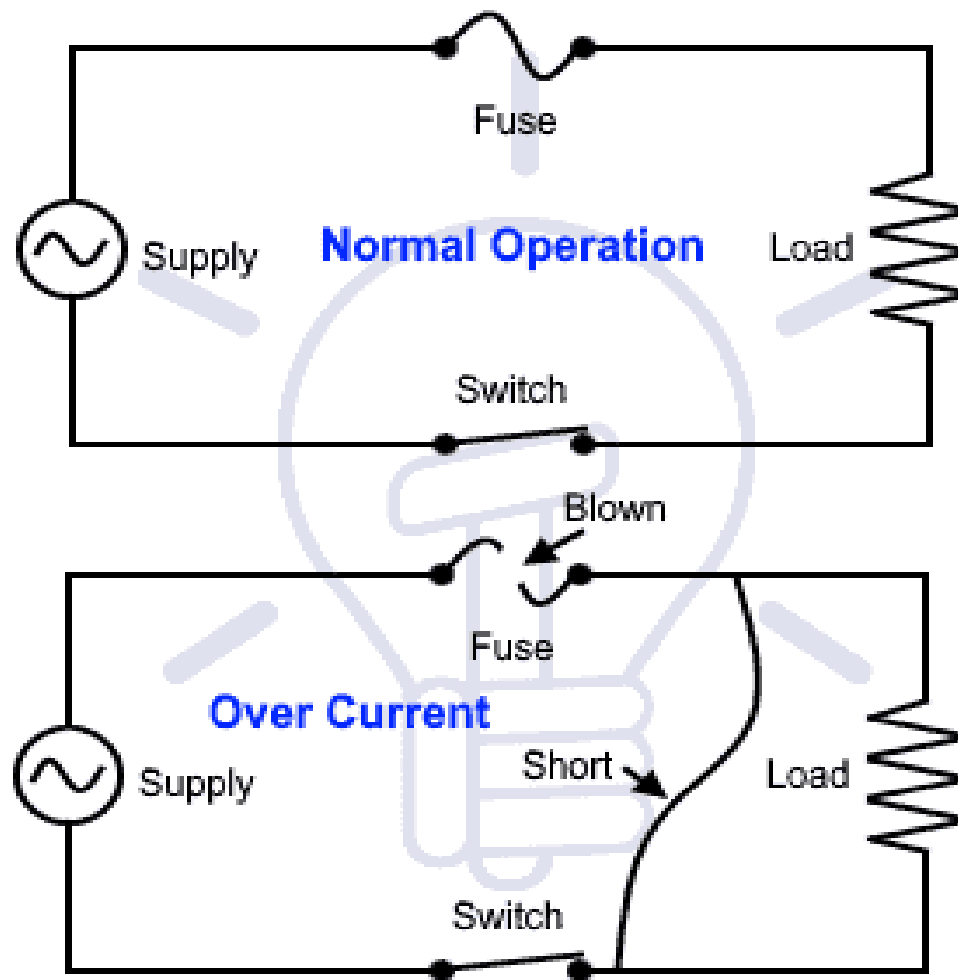


Different Symbols of Fuse

Construction & Working of a Fuse

- A general fuse consists of a low-resistance metallic wire enclosed in a non-combustible material. It is used to connect and install in series with a circuit and device that needs to be protected from short circuit and overcurrent; otherwise, an electrical appliance may be damaged in case of absence of a fuse and circuit breaker, as they are unable to handle the excessive current according to their rating limits.

The **working principle of a fuse** is based on the “Heating effect of current” i.e., whenever a short circuit, overcurrent, or mismatched load connection occurs, then the thin wire inside the fuse melts because of the heat generated by the heavy current flowing through it. Therefore, it disconnects the power supply from the connected system. In normal operation of the circuit, the fuse wire is just a very low resistance component and does not affect the normal operation of the system connected to the power supply.



Working of a Fuse

Characteristics of a Fuse

- Different types of fuses can be categorized on the following characteristics.
 - ❑ Current Rating & Current Carrying Capacity of Fuse
 - ❑ Voltage Rating of Fuse
 - ❑ Breaking Capacity of a Fuse
 - ❑ I^2t Value of Fuse
 - ❑ Response Characteristic
 - ❑ Packaging Size

Fuse Current Carrying Capacity

- Current carrying capacity is the amount of current that a fuse can easily conduct without interrupting the circuit.
- Breaking capacity:

The value of maximum current that can safely be interrupted by the Fuse is called Breaking Capacity and should be higher than the prospective short-circuit current.

Rated Voltage of Fuse

- Expect the current capacity of current; there is the maximum voltage rating a fuse can handle safely.
- Each fuse has a maximum allowed voltage rating.
- For example, if a fuse is designed for 32 volts, it cannot be used with 220 volts; different amounts of isolation are required in different fuses working on different voltage levels.
- Bases of voltage rating, a fuse can be HV (High Voltage) LV (Low Voltage), and Miniature Fuses.

I^2t Value of Fuse

- The I^2t terms related to the fuse are normally used in a short circuit condition.
- It is the amount of energy that carries the fuse element when the electrical fault is cleared by the fuse element.

Response Characteristic of a Fuse

- The speed at which the fuse blows depends on the amount of current flowing through its wire. The higher the current flowing through the wire, the faster the response time.
- The response characteristic shows the response time for an [overcurrent](#) event. Fuses that respond rapidly to the overcurrent situation are called ultra-fast fuses or Fast fuses. They are used in Many semiconductor devices because semiconductor devices are damaged by overcurrent very rapidly.
- There is another fuse which Is called a **slow-burning fuse**. Switch fuses do not respond rapidly to the overcurrent event, but blow after several seconds of overcurrent occurrence. Such fuses found their application in motor control electronics systems because [motors](#) take a lot more current at starting than running.

Packaging Size

- AC and DC fuses have a slightly different packaging type; in the same way, different applications require different packages to be used accurately in the circuit.

Classification of Fuses

- Fuses can be classified as “One Time Only Fuse”, “Resettable Fuse”, “Current limiting and non–current limiting fuses” based on the usage for different applications.
- One-time use fuses contain a metallic wire, which burns out when an overcurrent, overload, or mismatched load connection event occurs. The user has to manually replace these fuses.
- Switch fuses are cheap and widely used in almost all electronics and electrical systems.

The resettable fuse automatically resets after the operation when a fault occurs in the system.

In the current-limiting fuse, they produce high resistance for a very short period, while the non-current-limiting fuse produces an arc in case of high current flow interrupt and limit the current in related and connected circuits.

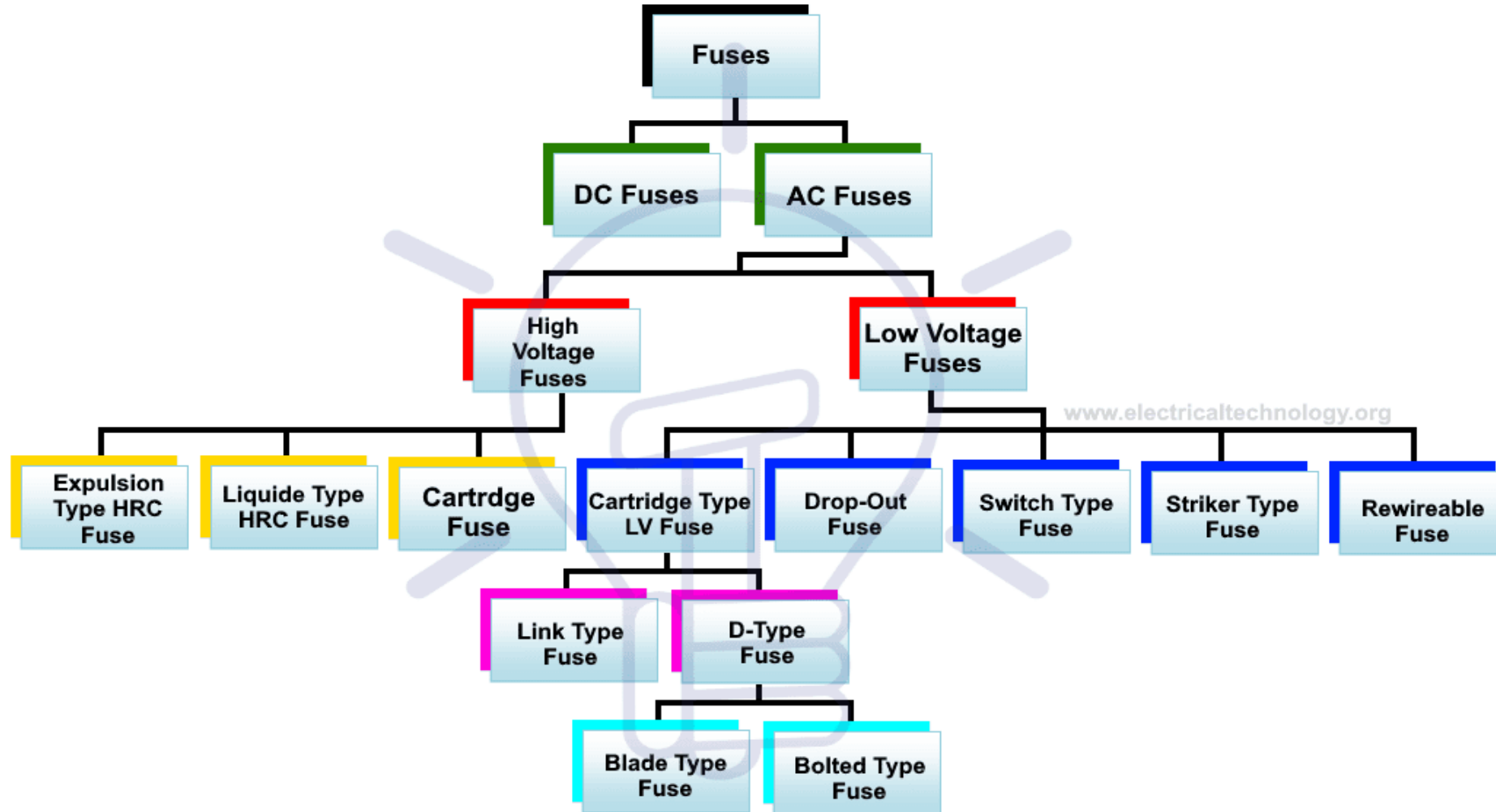
Types of Fuses



Different Types of Fuses

There are **different types of fuses** available in the market, and they can be categorized on the basis of Different aspects.

Fuses are used in AC as well as DC circuits.



Fuses can be divided into two main categories according to the type of input supply voltage.

- . AC fuses
- . DC fuses

-DC Fuses

In a DC system, when the metallic wire melts due to the heat generated by overcurrent, an arc is produced, and it is complicated to extinguish this arc because of the DC constant voltage.

So in order to minimize the fuse arcing, DC fuses are a little bigger than an AC fuse, which increases the distance between the electrodes to reduce the arc in the Fuse.

-AC Fuses

On the other hand, i.e., in the AC system, the voltage with 60Hz or 50Hz frequency changes its amplitude from zero to 60 times every second, so the arc can be extinct easily as compared to DC.

Therefore, AC fuses are a little bit smaller in sizes as compared to DC fuses.

Low voltage fuses

- **1- Rewirable Fuse**

It is made from a porcelain or ceramic insulator).



2- Cartridge Fuse.

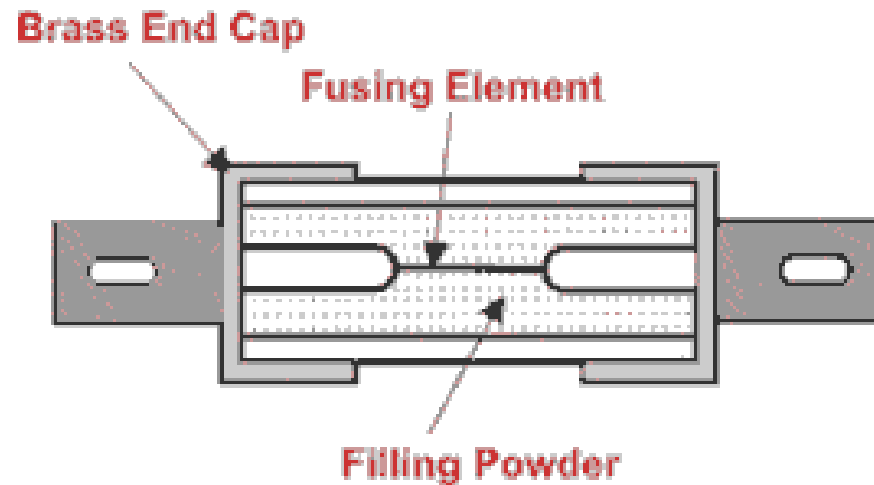
Advantage

- The wire is enclosed in a cartridge-type container.
- The wrong size of fuse cannot be fitted since it is with a different size for a different current.
- The fuse wire does not deteriorate and is more reliable in o



3- High Rupturing Capacity (H.R.C)

It is a cartridge-type with a silver element connected between two end-contacts of a ceramic tube filled with a special quartz powder.



Construction of HRC Fuse

Advantage:

- This type of fuse is very reliable in performance and does not deteriorate, and has a high speed
- When the fuse silver blows (which rapidly melts), produced with the filling powder, high-resistance materials are formed in of operation and the path of the arc, causing it to be extinguished.
- Ability to safely interrupt short circuit currents of much higher values (higher rupturing capacity).



Advantage

- ☐ Low cost
- ☐ Wire may be easily available.

Disadvantage

- ☐ The wrong size of wire to be fitted in the fuse cause wrong operation at high current circuitst which may be dangerous for the circuit protected and not adequate for electrical arc extinguish